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Design and development of a low-cost 5-DOF robotic arm for lightweight material handling and sorting applications: A case study for small manufacturing industries of Pakistan

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ABSTRACT

Due to the ever-increasing demand for higher production rates and the shortage of skilled labor in small industries, material handling and sorting have become extremely tedious and challenging. Industrial automation-led effective material-handling solutions like robotic arms have gained immense importance as they provide an alternative to human involvement, contribute to higher sorting accuracy, and provide enhanced safety. However, the adaptability of these robotic arms in small manufacturing industries in Pakistan is mainly hindered due to their higher costs and concerns with their structural durability. This paper presents the development of a low-cost 5-DOF robotic arm with a designed payload limit of 1 kg and automatically sorts objects fed through a conveyer belt. Catering to the compact sizing, high strength, and lower payload requirements of small industries, aluminum was selected as the material of the robotic arm due to its superior strength-to-weight ratio while being lightweight. Arm geometry was developed using SOLIDWORKS® software, which was further processed in ANSYS® software to perform the static structural analysis of the robotic arm using Finite Element Analysis (FEA). The fine meshing of the robotic arm assembly was done using triangular elements with the total number of elements and total nodes 52134 and 89104, respectively. A single point load was applied on the end effector, and the force was kept downward with an incremental loading of 1 kg starting from 100 g. These FEA simulations show that the robotic arm can hoist considerable weight while maintaining its structural integrity and directionality. The proposed robotic arm is also well-suited for manipulating objects in tight spaces due to its compact size and customizable range of motion, making it an ideal choice for applications that require precise manipulation of light loads.

1. Introduction

Material handling and sorting are necessary for many industrial operations; however, manual handling that uses human labor is a tedious task [1], especially when it involves a large volume of products. The inclusion of automation has enabled the manufacturing industry to meet the increased production rates, improve production efficiency, and reduce production costs [2]. From the perspective of industrial automation, effective material handling solutions like robotic units have gained immense importance as they are replacing human sorters for production line work [3–5]. Nowadays, robots are increasingly being deployed to accomplish industrial tasks with strict requirements of accuracy, precision, repeatability, mass production and quality, ease of human effort, and cost-effectiveness [6]. Robots deal with various operations such as design, operation, construction, controlling sensory feedback, and information processing for computer systems [7]. Advances in sensing, actuation, and computing systems have opened new horizons for robotics applications in a diverse range of fields such as

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biomedical, sports, space, and many more [8–10].

Generally, two major areas in which robots can be divided are industrial and service. A service robot is a robot that performs services beneficial to the well-being of humans and equipment, excluding manufacturing operations [11]. Industrial robots can perform tasks involving repetitiveness, accuracy, endurance, speed, and reliability [12]. The most commonly used industrial robot in manufacturing is the robotic arm which has many different applications in industry, such as painting, drilling, welding, thermal spraying, material handling, recycling, and disassembly [13,14]. A robotic arm is a robot manipulator wherein, links form a kinematic chain connected by joints that allow rotational or translational displacement [15]. The end-effector of the robotic arm defines the purpose of the arm and is capable of diverse functions, for example, grasping, manipulating, and pushing [16]. The availability of robotic arms for manufacturing industries will provide an alternative to human involvement, diminishes the probability of human error, and contribute to higher sorting accuracy and increased safety [17]. Adoptability of these robotic arms by local industries is hindered in Pakistan, due to their higher costs; however, research and development supported by several private and government-backed institutions have led the new ways [18].

Olawale et al. [19] developed a microcontroller-based robotic arm that could pick and release small objects through magnetizing and demagnetizing. 8051 microcontroller was interfaced with three stepper motors, each at the base, shoulder, and wrist of the anthropomorphic robot design. Sahoo et al. [20] designed and developed three degrees of freedom (DOF) robotic arm for medical surgery that use an Arduino microcontroller interfaced with stepper motors and a LabVIEW interface panel to control the rotation or movement of the arm with improved accuracy. A graphical simulator is designed to mimic the human arm simultaneously with the mechanical robot arm.

Elfasakhany et al. [21] proposed the low-cost design of four DOF robotic arm for less complex tasks such as light material handling. Thin acrylic sheets were used as lightweight and affordable material to manufacture the robotic arm, whereas the Arduino microcontroller and LabVIEW software was interfaced with servo motors at each joint to perform inverse kinematics control. The maximum payload of the developed robotic arm was 50 g with normal current consumption, which restricts its usage for industrial scenarios with higher payload requirements. Kruthika et al. [22] designed and developed five DOF articulated robotic arm as an assistive robot to feed aged and specially challenged people. The proposed design used an Arduino Mega 2560 microcontroller interfaced with one servo motor at the base joint and four DC-geared motors, each at the elbow, shoulder, wrist, and gripper joints. Aluminum was selected as the material to manufacture the robotic arm due to its strength, lightweight, and ease of machining. The kinematic model of the robotic arm from base to gripper is implemented using MATLAB. The actuators of each robotic arm joint (in control) were controlled through a Graphical User Interface (GUI) in MATLAB. Giannoccaro et al. [23] proposed a mechatronics application of a robotic arm that can sort different objects using Radio Frequency Identification (RFID) tags. The proposed system sort objects based on an integrated RFID and webcam to capture the position and identity of an object, solve the inverse kinematics for joint configuration and then transmit the information to each of the five servo motors to achieve the desired end-effector position. The limitation of having objects of the same size and shape carry out their features by the webcam and assigning an identification tag to each object restricts its practical applications in industrial scenarios.

Several researchers [24–26] have addressed the problem of material sorting using robotic arms based on different color-sensing modules. Khairuddin et al. [27] developed a color-sorting three-degree-of-freedom robotic arm that uses Arduino UNO microcontroller-based DC servo motor control. For color detection, a TCS3200 sensor converts the light intensity to frequency. Temperature, light intensity, and the distance of the object from the sensor significantly impact the accuracy of the designed system. Vision-based manipulators have recently gained much interest due to their ability to detect complex objects, enhanced sorting speeds, and increased sorting accuracy; however, they are prone to drop accuracy as lighting condition changes and requires computationally expensive image processing algorithms that need expensive hardware modules for their implementation [28–30].

All the above-discussed robots are usually driven by pneumatics, electrical, or hydraulics for actuating their bodies [31]. A soft robot offers safety, a natural adaptation to its environment, a motorless driven mechanism, and highly deformed structures, absorbs shocks, and protects it from mechanical impacts under intrinsic compliance [32]. Over to traditional rigid robots, soft robots can adapt to any environment for multi objects under hot and cryogenic conditions [33]. Recently, Li et al. [34] investigated the performance inflatable robotic arm (soft robot) for sensitive inspection. The proposed inflatable robotic arm demonstrated high compliance to practical applications with superior compliance to cluttered environments. However, researchers reported various issues with soft robotics, including the relatively low force exertion and control complexity. Several types of damage are highly vulnerable to soft robotics, such as low fatigue, interfacial debonding, overloads, cuts, tears, and perforations. Table 1 comprehensively summarizes the advantages and drawbacks of traditional and novel materials used as a robotic arm.

Finite Element Analysis (FEA) is a conventional and powerful tool for the design verifications of a system in its early stage that can lead to an optimized structure before the fabrication [42–44]. These FEA techniques permit the optimization, and automation, of the design, which is highly imperative for various products, structures, and shapes to improve their life cycle related to small industries perspective [45]. For

Table 1
Comparative analysis of various materials potentially employed for a robotic arm.

Material	Cost	Advantages	Constrained/Disadvantages	Ref.
Metals/Alloys	Cost-effective	• Can lift heavy weight. • Successfully employed in any production industry. • Embedded security features.	• Only detect large strains. • The heavyweight robotic arm. • Expensive if base metal/alloys are from nickel, chromium etc. • Wide working space is usually required.	[35,36]
Composites Materials	Moderate expensive	• The robotic arm is lightweight. • Can detect small strains. • Simple operation. Simple programming.	• The viscoelastic behavior of polymeric composites caused a delay in response. • Hysteresis	[37,38]
Soft Polymeric Materials (Soft robots)	Expensive	• Based on the entire motorless deformation mechanism. • Can detect much smaller strains. • Can lift toxic objects in extreme environmental conditions. • Can operate in limited space. • Flexibility of deployment.	• Relatively new technology has limited commercial applications. • Usually lift lightweight objects. • Programming under stimulant conditions is time-consuming.	[39–41]

a robotic arm manipulator to withstand heavy loads, appropriate materials should be selected which improve the load-bearing capacity of the robotic arm manipulators and enhance their lifespan while being lightweight. For instance, Marumo et al. [46] recently successfully employed FEA analysis to determine lightweight material such as carbon epoxy for a potential robotic arm material by comparing the equivalent stress and deformation results with conventional aluminum alloy. Furthermore, Chowdhry [47] used FEA simulations to find modal frequencies of the 3-DOF robotic arm utilized for structural optimization with low mass and high natural frequencies of the robotic arm. Shanker et al. [48] used ANSYS-based FEA simulations to find the optimum payload and to check structural deformations for a 3-DOF robotic arm under different loading conditions ranging up to 0–20 N. Kader et al. [49] used CATIA-based FEA simulations to optimize the structure of its designed 3-DOF robotic arm by identifying the critical stress areas and maximum when it carries out the pick and place operation. Kouritem et al. [50] explored a complete model analysis of 6-DOF robotic using ANSYS. Moreover, they calculated as well as optimized fundamental frequencies, modal shapes, and proper material selection mechanisms for safe frequency operation ranges for the robot arm. Insight of simulation results was promising for the highly improved factor of safety of design with low material cost.

Maximizing the advantages of industrial automation for material handling and sorting tasks in small manufacturing industries requires robotic arms that have low cost, higher DOF, compact sizing, high strength, and small payload requirements. The literature lacks an FE-based structural analysis of a 5-DOF robotic arm that can be used to verify the design requirements of the payload. To address the shortcomings identified in the above-reviewed literature, this paper proposes the design of a low-cost five-DOF robotic arm that uses Arduino microcontroller-based servo motor control for automatically sorting objects fed through a conveyer belt. Furthermore, the FEA-based structural analysis of the designed robotic arm is carried out to check the total deformation and stresses due to the payload.

2. Methodology

As depicted in Fig. 1 through a schematic diagram, the designed material sorting application of the robotic arm. For instance, a specific material with dimensions similar to or less than the dimension of a robotic arm will place on the conveyor belt, and it will move along the belt and pass through a sensing chamber; it will detect the metal or non-metal object and pass the information through the robotic arm. The later robotic arm will pick the object and place the object either on the left or right side based on the metal or non-metal material. While Fig. 2 depicts the detailed working of the designed robotic arm to segregate materials fed through a conveyer belt. The presence of the object on the conveyer is first sensed using an Infrared (IR) sensor, which starts the gear motor of the conveyer belt. The controlling circuit of the conveyer consists of a potentiometer that allows the operator to manually regulate the conveyor belt speed via the regulation knob. When the object reaches the magnetic sensing unit, it detects the object and sends a signal to Arduino Microcontroller that activates the robotic arm. If the object detected on the conveyer is metal, Arduino-controlled servo motors at each joint of the robotic arm work according to the preset kinematic model and place that box on the left side of the conveyor. If the object detected on the conveyer is non-metal, Arduino-controlled servo motors at each joint of the robotic arm work, according to the preset kinematic model and place that box on the right side of the conveyor. After placing the object on its desired point, the robotic arm comes to its preset zero position before the execution of the next cycle of sorting.

To accurately execute the functioning of the designed robotic arm, the working of several electrical and electronic components that drive the mechanical assembly is essentially required. A transformer thus enables an alternating current (AC) voltage to be "stepped up" by making Ns larger than Np or "stepped down" by lowering Ns than Np. 230/220 V power source is converted from the main supply via a 12-0-12 V step-down tap transformer. This AC power has to be converted to direct current (DC) to be supplied to controllers and DC motors. Full-wave rectification transforms the input waveform polarities to DC and is more efficient. 6A diodes are used for the rectification of voltage. The AC voltage is thus now transformed to the pulsing DC. A smoothing circuit or filter is needed to provide a stable DC from a rectified AC source. This pulsation is eliminated via the 1000 μF condenser filter circuit. A kind of relay that can manage the power needed to operate an electric motor directly is termed a contractor. Solid-state relays regulate power circuits with no moveable components and use a semiconductor to switch. Fig. 3 shows the description of various electronic components used in controlling circuitry.

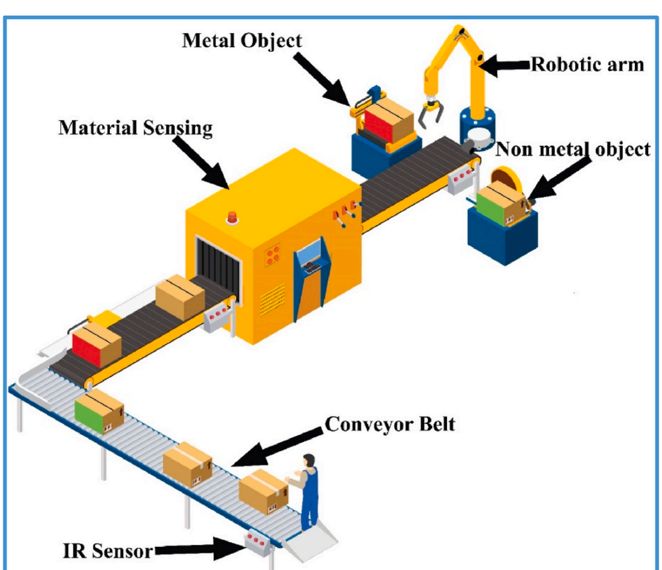

Fig. 1. Schematic of the designed material sensing and sorting system.

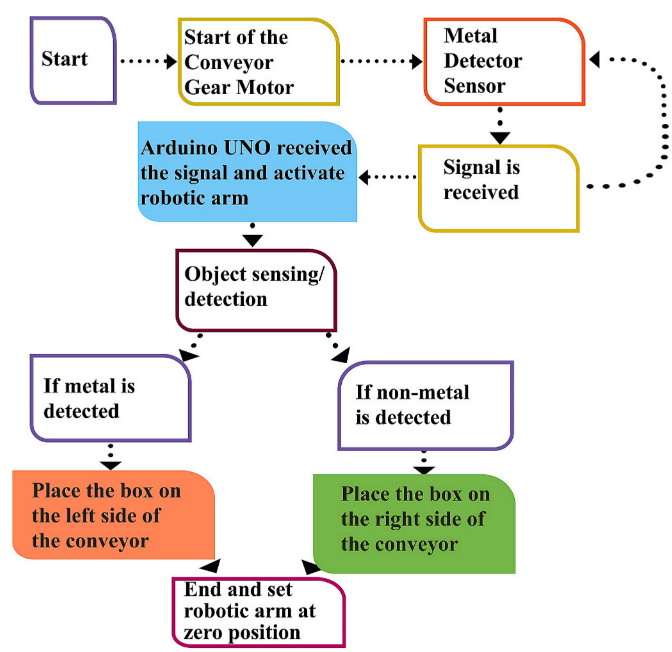

Fig. 2. Flow chart describing the detailed working of the developed robotic arm.

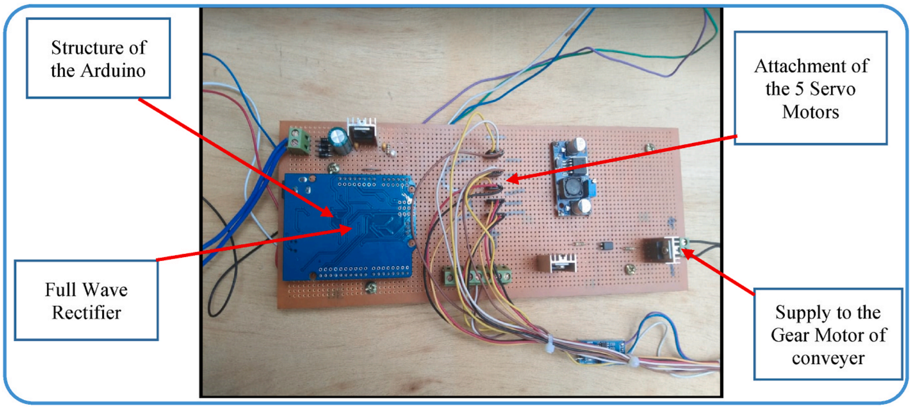

Fig. 3. Description of various electronic components used in controlling circuitry.

Table 2
Motor size defining.

S/N	Joint name	Total resistance torque (kg. cm)	Servo motor specifications	Servo motor name
1	Grip joint		Torque of 1.8 kg/cm at 4.8 V	Tower proSG 90
2	Wrist joint	$T4g + T4i = 6.012 \times 10^{-4}$	Torque of 9.4 kg/cm at 4.8 V	TowerproMG 995
3	Elbow joint	$T3g + T3i = 2.756$	Torque of 9.4 kg/cm at 4.8 V	TowerproMG 995
4	Shoulder joint	$T2g + T2i = 6.055$	Torque of 9.4 kg/cm at 4.8 V	TowerproMG 995
5	Base joint	$T1g + T1i = 0.3040$	Torque of 9.4 kg/cm at 4.8 V	Towerpro MG995

2.1. Magnetic sensing circuit

We used a condenser loaded with the rising pulse or spike, and it took a few pulses to charge the condenser to the extent where Arduino analog pin A5 reads its voltage. Arduino reads this condenser's voltage using ADC. After sensing the voltage, the condenser is rapidly drained, and the cabin is set to low. It takes around 200 μs to finish this entire procedure. We repeat measurements and take an average of the findings for better outcomes. This is how the estimated coil inductance may be measured. After the result, we transmit the results to the light-emitting diode (LED) and buzzer to identify metal presence.

2.2. Servo motors selection

For the development of the designed robotic arm, servo motors were used because of their ability to control precisely the velocity, angular position, and acceleration of the output shaft. The formulae established under the torque estimation section and the inertial resistive torque equation were used to evaluate resistive torque acting at each joint. A servo motor with appropriate torque and voltage rating was then selected for each joint. The servo motor rating is unique since torque resistive value is unique to a joint. Table 2 shows total resistive torque and servo motor specifications per joint in a robotic arm.

2.3. Microcontroller interfacing

A microcontroller is considered as a whole computer manufactured on a single chip having a high concentration of on-chip facilities, including parallel input/output ports, serial ports, timers, interrupt control, analog-to-digital converters, counters, random access memory, and dedicated read-only memory [51–53].

The microcontroller Atmega328 on the Arduino board is a high-performance Atmel 8-bit Reduced Instruction Set Computer (RISC) based microcontroller. It combines 23 general-purpose I/O lines, 32 general-purpose working registers, three flexible timer/counters with compare modes, internal and external interrupts, a 6-channel 10-bit A/D converter, a programmable watchdog timer with an internal oscillator. The device operates between 1.8 and 5.5 V. Arduino UNO board was selected as the main interfacing device for the designed robotic arm due to its operating speed, the number of accessible I/O pins, ease of development facility and availability, and low price compared to other microcontrollers. Fig. 4 depicts the schematic diagram of Arduino UNO interfacing with the developed robotic arm assembly.

2.4. Mechanical design of the 5-DOF robotic arm

The kinematic dimensions of the robotic arm are $l_1 = 41.04$ mm, $l_2 = 93.55$ mm, $l_3 = 65.115$ mm and the $l_4 = 59.9$ mm and $d = 28$ mm. Table 3 describes the details of all specifications for designing the proposed robotic arm. The joints of the robotic arm are driven by 5.17: 1 planetary gearbox servo motors, which can produce a maximum torque of 9.2 Nm/cm, resulting in a maximum payload of 1 kg for the robotic arm. Moreover, it requires a maximum of 10 W of power to operate. This high power-to-load ratio makes it suitable for carrying out complex tasks efficiently and effectively. It is also easy to use and maintain due to its simple design, providing reliable performance in the long run. The robotic arm is also well-suited for manipulating objects in tight spaces, due to its compact size and adjustable range of motion. All these features make it an ideal choice for applications that require precise and accurate

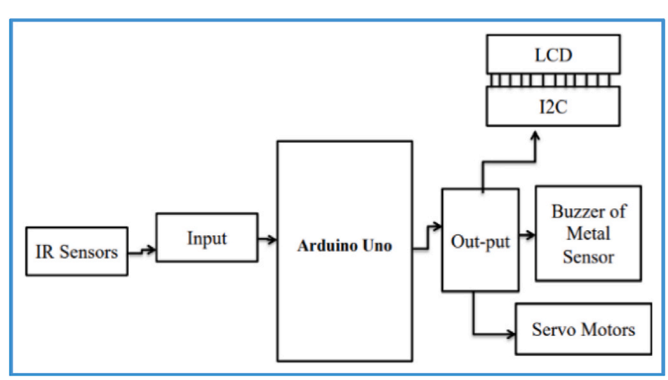

Fig. 4. Schematic diagram of Arduino UNO interfacing with developed robotic arm assembly.

Table 3
Specifications of the designed robotic arm.

Specification	**Number of the Axis**	**5 Axis**
	Payload	1 kg
	Power supply	110 V/220 V, 50/60 Hz
	Power in	12 V/3 A
Axis Movement	**Axis**	**Range**
	Joint 1 base	θ_1
	Joint 2	$90^\circ - \theta_2$
	Joint 3	θ_3
	Joint 4	$180^\circ - \theta_4$
	Joint 5	180°
Physical	Weight	2.9 kg
	Dimenssion	200 mm × 150 mm
	Materials	Aluminum
	Controller	Arduino

manipulation of light loads.

In the robotic arm's joints, five servo motors have been installed. These motors enable it to rotate to five degrees. The clipper distance it can reach with its jaws is 55 mm. The various Computer Aided Design (CAD)-based separate models of robotic arms are depicted in Fig. 5, whereas Fig. 6 shows the complete assembled part of the robotic arm in CAD.

2.5. Material selection

The construction of a robotic manipulator needs a selection of material that fulfills certain aspects of the design criteria. Durability, strength, low weight, easy availability, and low price were the main aspects considered when selecting the material for the design and development of this robotic arm. Material with enough strength was utilized to avoid the failure of every arm connection when the load was applied. Aluminum, the lightweight and high-strength material, was selected as the material of the robotic arm to ensure that it can withstand the designed payload limit without failure, resist corrosion, and ease of machining since certain components of the robotic arm needed the cutting of complex forms.

3. Results and discussion

3.1. Forward kinematic modelling

The forward kinematics calculates the orientation and position of the end-effector to the given reference frame, given the set of joint-link parameters. The forward kinematics of the robotic arm was established by the Denavit-Hartenberg (D-H) serial link settings [54] which are a systematic description and representation of the spatial geometry of the elements of a kinematic chain relative to a fixed reference system [55]. The geometrical configuration of the designed robotic arm joints comprises a base, elbow, shoulder, wrist, and gripper. Four parameters of the robotic arm, such as link length, link twist, link offset, and joint angle (α_i, a_i, d_i, and θ_i) are used for D-H parameters work. Orthonormal coordinate frames are attached to each of the links. Link length α_i,

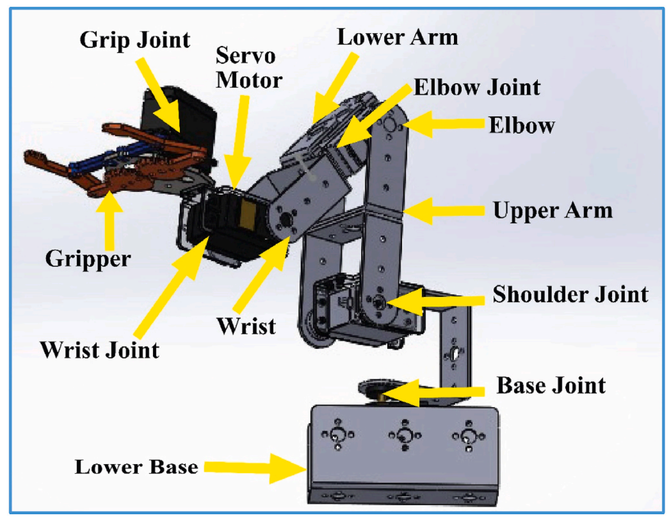

Fig. 6. Description of the designed robotic arm model.

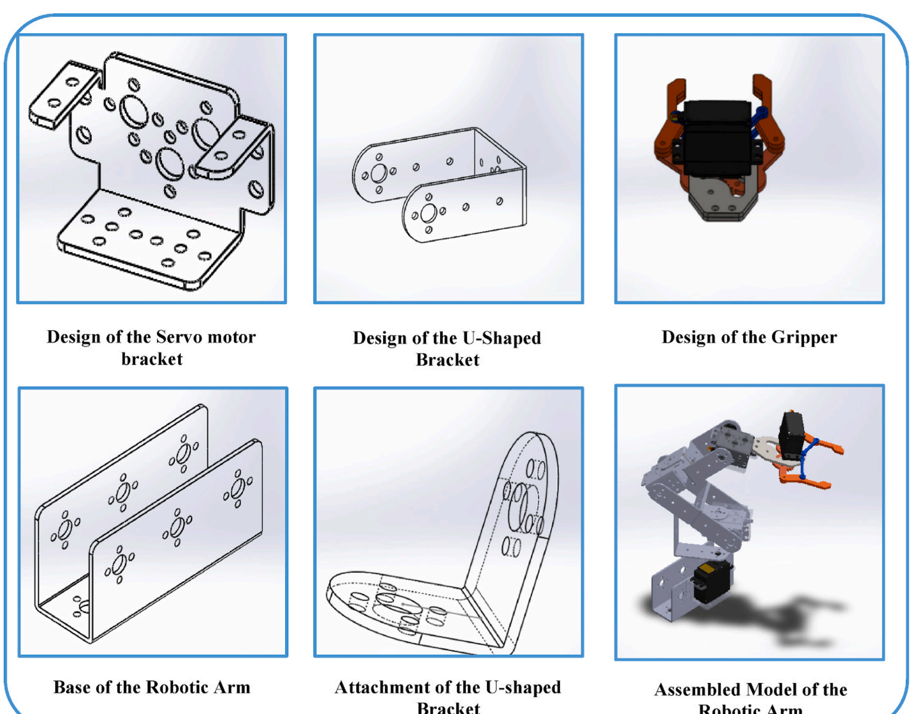

Fig. 5. CAD models of the different components of the robotic arm assembly.

defined as the distance between two successive joint axes z_i and z_{i-1} measured along the common normal. Link twist a_i, defined as the twist angle between joint axes z_i and z_{i-1} measured about x_{i-1}. Link offset d_i, defined as the distance between axes x_i and x_{i-1} measured along z_i. Joint angle θ_i, the angle between axes x_i and x_{i-1} measured about z_i. Fig. 7 illustrates the schematic diagram of the chosen axes.

Complete D-H parameters are discussed in Table 3 that is formed by the following method:

for $\boldsymbol{i} = \boldsymbol{1}$:

$\alpha_i\text{-}1 = \alpha_0 = 0^\circ$, that is the angle from z_0 to z_1 measured along x_0
$a_{i-1} = a_0 = 0$ mm, that is the distance from z_0 to z_1 measured about x_0
$d_i = d_1 = 0$ mm, that is the distance from x_0 to x_1 measured along z_1
$\theta_i = \theta_1$, that is the angle from x_0 to x_1 measured about z_1

Similarly, for i = 2, 3,4, and 5, calculations are performed and Table 4 is constructed.

3.2. Finite Element Analysis of the designed robotic arm

The structural analysis of the robotic arm under a particular loading is performed using a Finite Element (FE) model. Arm geometry has been developed using SOLIDWORKS® software, which is converted into IGS file format. This file was imported into ANSYS® software, and component deformations and stresses are calculated by using the static structural tool. During the FEA simulation, various boundary conditions like nodes, elements, loads, and material properties were defined to model the mechanical behavior of the robotic arm accurately. The mesh size of elements is also an important factor in FEA analysis as it affects the accuracy of the results. The fine meshing of the robotic arm was done by using triangular elements. The total number of elements and total nodes were 52134 and 89104, respectively. In addition, a single point load was applied on the end effector of the robotic arm, and the force was kept downward.

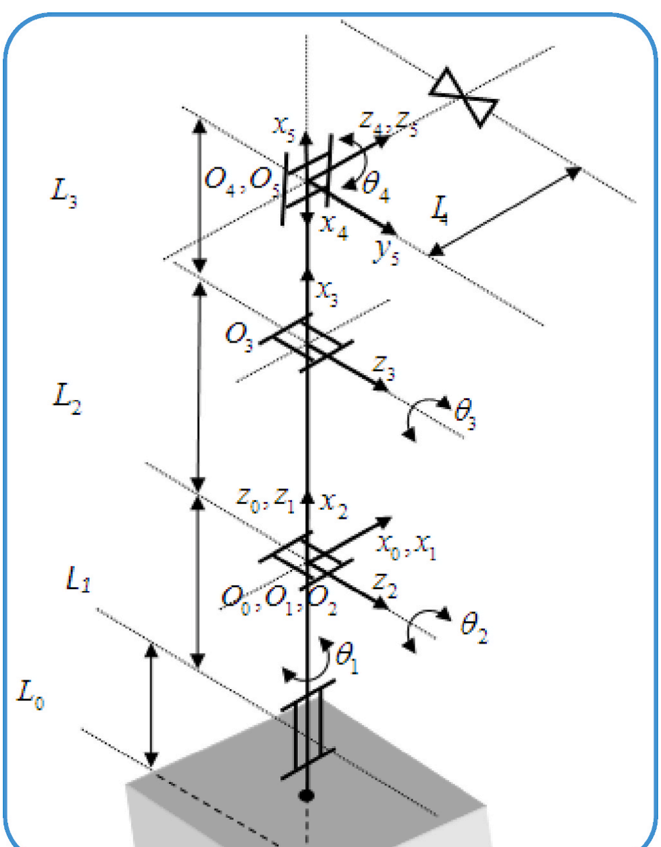

Fig. 7. Schematic diagram of the chosen axes [56].

Table 4
D-H Parameters of the designed robotic arm.

No of joints	α_{i-1}	a_{i-1}	d_i	θ_i
1	0°	0	0	θ_1
2	90°	0	0	$90^\circ - \theta_2$
3	0°	L_2	0	θ_3
4	90°	L_3	0	$180^\circ - \theta_4$
5	0°	0	0	180°

Initially, a load of 100 g was applied to the robotic arm to observe any structural changes. The results showed that the robotic arm could bear the load without any noticeable changes in the overall deformation. The directional deformation of the robotic arm had also been observed, and it was seen that the slight changes in the structure did not compromise the structural integrity of the arm. When the load of 400 g was applied to the model, a minor deformation amplitude was observed, further confirming the viability of the designed model at increased payloads. The deformation was more pronounced when the load was increased to 1 kg, which falls in line with its designed payload range of 1 kg. The observations from these FEA simulations have shown that the robotic arm can hoist considerable amounts of weight while maintaining its structural integrity and directionality. The von Mises stress, which measures the total amount of stress in the model, remained constant throughout the analysis as the model can withstand the increased load without any damage or deformations that could lead to failure and did not engender any consequential impacts. Fig. 8 presents various FEA-based results for the 5-DOF robotic arm.

3.3. Fabricated model

Fig. 9(a)-9(b) illustrate the real and stationary positions of the robotic arm, respectively, before the execution of the command from the microcontroller. The robotic arm will be in this position until there shall be no detection of the object, whether metal or non-metal. Fig. 9(c) represents the real-time capture of an image of the robotic arm when capturing the non-metals is being proceeded. In this case, first, the robotic is stationary when the metal passes through the white box; the microcontroller sends the coded message to the robotic then the actuator clips the non-metal and places it on the right side. Fig. 9(d) shows the real-time image of the robotic arm when it clips the metal object. The same procedure for non-metal is carried out. The only difference is robotic arm placed the carried metal on the left side of the specific place. Thus, placing the metal on the left side of the place and the same action can proceed. Finally, the robotic arm will be at zero and again carry out the action depending on the type of metal placed on the conveyor belt.

The accuracy of the designed system depends upon the performance parameters: True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN), derived from the testing dataset. A total of 143 objects were used in the testing phase to check the efficiency of the developed robotic arm for material sorting applications. Out of 143 tested objects, $TP = 61$ were originally metals but sensed and placed by the robotic arm as metals, $FP = 11$ were originally metals but sensed and placed by the robotic arm as non-metals, $FN = 12$ that were originally non-metals but sensed and placed by the robotic arm as metals and $TN = 59$ that were originally non-metals but sensed and placed by the robotic arms as non-metals. The following formulae may be used to compute these parameters.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \qquad \text{Eq. (1)}$$

$\text{Accuracy} = 83.91\%$

We envisaged that a smart robotic arm has the potential to use in many applications, such as the smart robotic arm for material sorting

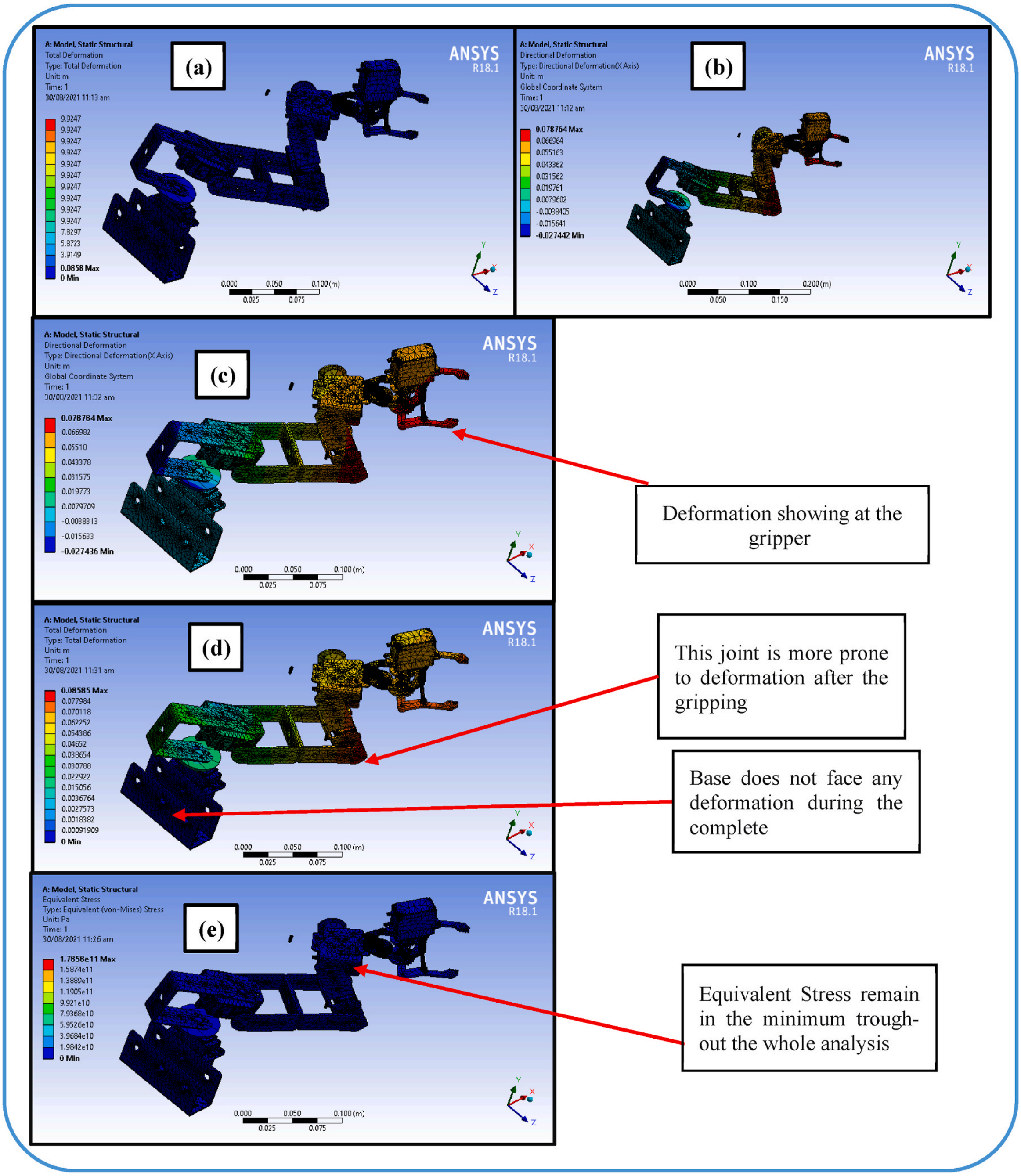

Fig. 8. Results of FEA on Designed 5-DOF Robotic Arm (a) Total deformation at 100g weight; (b) Directional deformation at 100 g; (c) Directional deformation results at 1 kg load; (d) Total deformation results at 1 kg load; (e) Equivalent stress.

applications as well as the movement of many heavy, hazardous, and bulky objects.

3.4. Cost analysis

We proposed a low-cost 5-DOF industrial robotic arm that is feasible for material sorting applications and can be a cost-effective solution for small industries to adopt the basic level of automation. Furthermore, the developed robotic arm has the potential to help the manufacturers by lowering labor costs as well as addressing the labor shortages for undesirable jobs such as where toxic substances handling is required, improving overall productivity and competitiveness. Table 5 reports the real-time cost analysis of the developed robotic arm.

4. Conclusion

Robotic arms are increasingly being deployed to achieve higher accuracy standards, repeatability, and ease of human effort in material handling and sorting tasks. Small industries have strict requirements of lower costs, structural durability, compact sizing, and lower payload requirements for the adaptability of these robotic arms in Pakistan. To serve the above purpose, the design and development of a low-cost 5-DOF robotic arm was proposed. Aluminum was selected as the material of the robotic arm due to its superior strength-to-weight ratio while being lightweight. Arduino UNO board was selected as the main interfacing device for the designed robotic arm due to its operating speed, the number of accessible I/O pins, ease of development facility, availability, and low price compared to other microcontrollers. Arduino-controlled

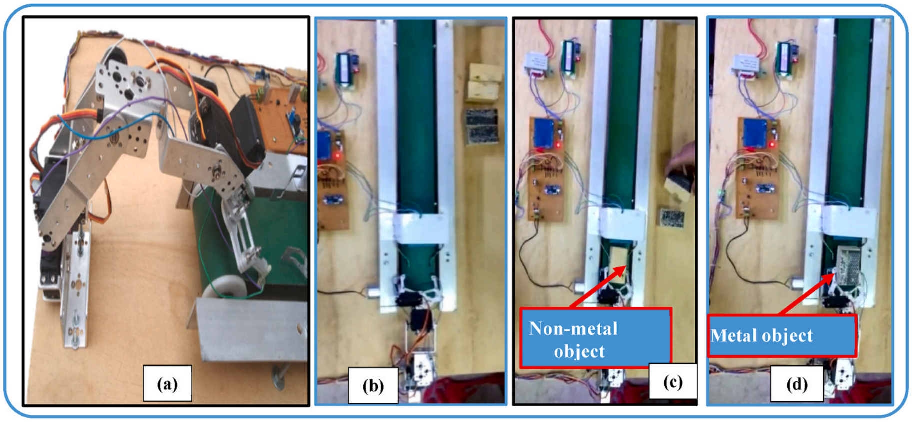

Fig. 9. (a) Real-time depiction of robotic arm working (b) Rest position (c) Sorting out the non-metal object and placing it on the right side (d) Sorting the metal object and placing it on the left side.

Table 5
Cost estimation of the developed robotic arm and conveyer assembly.

Part Name	Unit Price (PKR)	Total Price (PKR)
Railing	1000 Per Feet	6000
Rollers	250	1000
Belt	400	2400
LCD + I2C	–	500
Step down transformer	500	500
Arduino Uno Board	–	3500
Wiring	–	1000
IR sensors	–	200
Magnetic sensor + buzzer	–	500
Robotic arm material (aluminum)	–	8000
Gear motor	800	800
Servo motors	1200	6000
Fabrication cost	–	3000
Total cost in PKR	–	**33600**
Total Cost in USD	–	**~ 120**

servo motors at each joint of the robotic arm work according to the preset kinematic model and place the fed box on the left or right side of the conveyer based on whether the detected object is metal or non-metal. Cost analysis reveals that the complete material and fabrication cost of our proposed 5-DOF robotic arm with conveyer assembly is 120 *USD*. The major portion of this total cost comprises the material cost for the robotic arm structure, the costs of servo motors, and the cost of the Arduino UNO board. The developed robotic arm is highly feasible for small industries with payload requirements of up to 1 *kg*.

The current limitation of the proposed study is that it deals with a 5-DOF robotic arm with a maximum load capacity of 1 *kg* with a maximum dimension of the handling box to be 5 cm^2. Higher DOF robotic arms can be designed for more complex tasks by customizing the dimensions of link lengths, adjusting the size of the gripper, and selecting motors according to the torque requirements set by the intended payload. Furthermore, we have not studied the feasibility of developing a robotic arm for handling hazardous materials. We envisaged that the boom of soft robotics fabricated through advanced additive manufacturing routes would ultimately solve many problems, such as the heavy weight of grippers, providing motorless mechanisms for grasping and releasing various objects (more toxic and hazardous) more accurately and in a controlled manner.

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Conceptualization: Zain Ali, Muhammad Fahad Sheikh, Muhammad Yasir Khalid; Methodology: Zain Ali, Muhammad Fahad Sheikh, and Zia Ullah Arif; Software: Zain Ali, Muhammad Fahad Sheikh, Zia Ullah Arif; Formal analysis: Muhammad Yasir Khalid and Zia Ullah Arif; Investigation: Zain Ali, and Muhammad Fahad Sheikh; Writing – original draft preparation: Zain Ali, Muhammad Fahad Sheikh, and Muhammad Yasir Khalid; Writing – review & editing: Ans Al Rahsid, Rehan Umer and Muammer Koc; Supervision: Muhammad Fahad Sheikh, Rehan Umer and Muammer koc; Project administration: Ans Al Rashid, Rehan Umer.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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